

AN IBL INTRODUCTION TO GEOMETRIES



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An IBL Introduction to Geometries (Mark Fitch)

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Licensing

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CHAPTER OVERVIEW

1: Completeness and Consistency

[1.1: Finite Geometries](#)

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1.1: Finite Geometries

Definition: Intersect

Two lines **intersect** if and only if they share a point.

Definition: Parallel

Two lines are **parallel** if and only if they do not intersect.

Definition: Four Point Geometry

The four point geometry is defined by the following axioms and definitions.

1. There exist exactly four points.
2. Any two distinct points have exactly one line on both of them.
3. Each line is on exactly two points.

Explore the four point geometry as follows.

1. Draw and label four points.
2. Use axiom 2 to draw as many lines as possible.
3. How many lines exist in this geometry?
4. Find a pair of parallel lines.
5. Can you find three lines that are pairwise parallel?
6. Can you find a point that is on three lines?

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1.2: Completeness

Definition: Fano Geometry

Use the following axioms and definitions of intersection and parallel as a definition of the Fano geometry.

more stuff

1. There exists at least one line.
2. There are exactly three points on every line.
3. Not all points are on the same line.

Explore the Fano geometry as follows.

1. Draw a line using Geogebra.
2. Add a third point to the line.
3. Note that Axiom 3 requires one more point. Draw one.
4. Must any more lines be added? If so, do so.
5. How many lines are in this geometry?
6. Add the axiom: every point is on at least one line.
7. To your line with three points, and one point not on that line add any lines required by this new axiom.
8. Make sure these lines satisfy Axiom 2.
9. How many lines are in this geometry?
10. May any more lines be added? If so, do so, and be sure the axioms are satisfied.
11. How many lines are in this geometry?
12. What do the answers to 5, 9, and 11 say about this attempt at constructing a geometry?
13. What is needed to fix the difficulty noted in the previous question?

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1.3: Consistency

Definition: Five Point Geometry

Use the following axioms and definitions of intersection and parallel as a definition of the five point geometry.

1. There exist exactly five points.
2. There exist exactly five lines.
3. Any two distinct points have exactly one line on both of them.
4. Each line is on exactly two points.

Explore the five point geometry as follows.

1. Draw five points using Geogebra.
2. Use Axiom 3 to draw all required lines.
3. How many lines did you construct?
4. Compare this answer to Axiom 2.

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1.4: Explain and Apply

1. Explain what completeness is and why it is important. Note, an explanation is not a proof, just a clarification for someone else.
 2. Explain what consistency is and why it is important.
 3. Modify the axioms for the Fano geometry so they are complete and consistent. You might use the Four Point Geometry axioms as a model.
 4. Try to construct a Fano geometry with exactly seven points.
 5. Modify the axioms of the Five Point Geometry to be complete and consistent. You might use the Four Point Geometry axioms as a model.
 6. How many lines are there in this geometry?
 7. Modify the axioms of the Five Point Geometry to be complete and consistent and so that they result in five lines.
-

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CHAPTER OVERVIEW

2: Neutral Geometry

2.1: Triangles

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2.1: Triangles

2.1.1 Basic Triangle Theorems

Note all theorems in this section can and should be proved without using the parallel postulate.

Definition: Vertical Angles

The opposing angles formed by the intersection of two lines are called **vertical angles**.

Definition: Congruent Angles

Two angles are *congruent* ($\angle ABC \cong \angle DEF$) if and only if their measures are equal ($m\angle ABC \cong m\angle DEF$).

Theorem: Vertical Angle Congruence

Vertical angles are congruent.

A-B-C means that the points A, B, and C are colinear and B is between A and C.

Theorem: Pasch's Axiom

If a line ℓ intersects a triangle $\triangle ABC$ at a point D such that A-D-B then ℓ must intersect AC or BC.

Theorem: Crossbar

If X is a point in the interior of $\triangle ABC$ then ray AX intersects BC at a point D such that B-D-C.

Definition: Congruent Line Segments

Two line segments are congruent ($AB \cong CD$) if and only if their measures (length) are equal ($|AB| = |CD|$).

Definition: Isosceles

A triangle is **isosceles** if and only if two sides are congruent.

Theorem: Isosceles Triangle

In an isosceles triangle the angles opposite the equal sides are congruent.

Theorem: Perpendicular Bisector

A point is on the perpendicular bisector of a segment if and only if it is equidistant from the endpoints.

Definition: Exterior Angle

The supplementary angle formed by extending one side of a triangle is called an **exterior angle**.

Theorem: Exterior Angle

The measure of an exterior angle of a triangle is greater than the measure of either of the opposing angles of the triangle.

Definition: Congruent Triangles

Two triangles are congruent if and only if all their sides and angles are congruent ($\triangle ABC \cong \triangle DEF$).

2.1.2 Triangle Congruence Theorems

Determine if two triangles with two congruent sides and a congruent angle not between the two sides are congruent.

Theorem: Angle-Side-Angle

Two triangles are congruent if and only if two corresponding angles and the side between them are congruent.

Theorem: Angle-Angle-Side

Two triangles are congruent if and only if two corresponding angles and a side not between them are congruent.

Theorem: Side-Side-Side

Two triangles are congruent if and only if all three corresponding sides are congruent.

Theorem: Right Angle-Side-Side

Two right triangles are congruent if and only if two corresponding sides and a right angle not between those sides are congruent.

Theorem: Converse of Isosceles Triangle

If two angles of a triangle are congruent then the sides opposite those angles are congruent.

Theorem: Extended Inverse of Isosceles Triangle

If two sides of a triangle are not congruent then the angles opposite those sides are not congruent. Further the larger angle is opposite the longer side.

Theorem: Triangle Inequality

The sum of the lengths of any two sides of a triangle is larger than the length of the other side.

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3: Synthetic Euclidean Geometry

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3.1: Equivalent Parallel Postulates

Each of the following is an equivalent Euclidean postulate.

Equivalent Euclidean Postulates:

- (Playfair) Given a line and a point not on that line, there exists exactly one line through that point parallel to the given line.
- (Equidistance) Lines that are parallel are everywhere equidistant.
- (Euclid) Given two lines and a transversal of those lines, if the sum of the angles on one side of the transversal is less than two right angles then the lines meet on that side.

3.1.1 Preparation

These theorems do not require a parallel postulate.

Theorem: Alternate Interior Angles

If the alternate interior angles formed by a transversal of two lines are equal, then the lines are parallel.

Theorem: Point to line distance

The (shortest) distance between a point and a line is from the point to the foot of the perpendicular.

Theorem: Convenient Euclid Parallel Axiom

Given two lines and a transversal of those lines, if the sum of the angles on one side of the transversal is equal to two right angles then the lines are parallel.

3.1.2 Equivalency

The following theorem produces an easier to use version of Euclid's postulate.

Theorem

Euclid's postulate and Theorem  **Convenient Euclid Parallel Axiom** are equivalent to the following. "The sum of the angles on one side of a transversal of two lines is equal to the sum of two right angles if and only if the lines are parallel."

Theorem

Euclid's postulate implies Playfair's axiom.

Theorem

Playfair's axiom implies the alternate interior angle converse theorem.

The alternate interior angle converse theorem states Given parallel lines and a transversal of those lines, the alternate interior angles formed by the transversal are congruent."

Theorem

Playfair and the alternating interior angle converse theorem imply the equidistance of parallel lines.

Theorem

The equidistance of parallel lines implies Euclid's postulate.

 Theorem

Prove the three postulates are all equivalent.

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3.2: Similarity

3.2.1 Preparation Theorems

For $\triangle ABC$ construct line ℓ such that $\ell \parallel AC$ and B is on ℓ . What is the relationship between the three angles at (smaller than a straight angle) to the angles of the triangle?

Theorem: Triangle Angle Sum

The angle sum of the interior angles of a triangle is π .

Corollary: Euclidean Exterior Angle

The measure of an exterior angle of a triangle is equal to the sum of the two opposing, interior angles.

Definition: Parallelogram

A quadrilateral is a **parallelogram** if and only if both opposing pairs of sides are parallel.

Theorem

Opposite sides of a parallelogram are congruent.

Theorem

If a transversal intersects three parallel lines in such a way as to divide itself into congruent segments, then any transversal of these parallel lines is also divided into congruent segments.

3.2.2 Explore Similarity Theorems

Definition: Altitude

A line segment is an **altitude** if it connects a vertex of a triangle to the foot of the perpendicular on the opposite side.

Construct a triangle and enough parallel lines to divide one side of the triangle into four equal parts. Into how many parts do these lines divide the other sides?

Definition: Triangle Area

The area of a triangle is equal to one half of the product of one side times the length of the altitude from the opposing vertex to that side.

Construct $\triangle ABC$. $m \angle A > 90^\circ$. Construct DE such that $B-D-A$, $B-E-C$ and $DE \parallel AC$.

1. Construct AE and EF such that F is the foot of the perpendicular from E . Reduce the ratio of the areas of $\triangle DEB$ and $\triangle AED$.
2. Construct DC and DG such that G is the foot of the perpendicular from D . Reduce the ratio of the areas of $\triangle DEB$ and $\triangle CDE$.
3. Prove area of $\triangle AED$ is equal to the area of $\triangle CDE$.

3.2.3 Similarity Theorems

Definition: Similar Triangles

Two triangles are *similar* if and only if corresponding angles are congruent and the ratio of corresponding sides is constant.

 Definition

A line parallel to one side of a triangle and intersecting the other two sides divides those sides proportionally.

 Definition

If a line parallel to one side of a triangle divides the other sides proportionally then the two triangles (large and part) are similar.

 Theorem: Angle-Angle-Angle (AAA) Similarity

Two triangles are similar if and only if they have three congruent angles.

3.2.4 Extending Similarity

Construct a definition for similar quadrilaterals. Construct examples to show that your definition works.

Explain why similarity is not defined simply as "all angles are congruent."

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3.3: Concurrent

Definition: Median

A line is a **median** if and only if it connects a vertex of a triangle to the midpoint of the opposing side.

3.3.1 Explore

Geogebra will be helpful for performing these experiments. Be as detailed as you can with your conjectures.

Use the Geogebra example in 3.3.1 to experiment with the relationship of the three perpendicular bisectors of a triangle. Move the vertices of the triangle around. What remains true about the perpendicular bisectors?

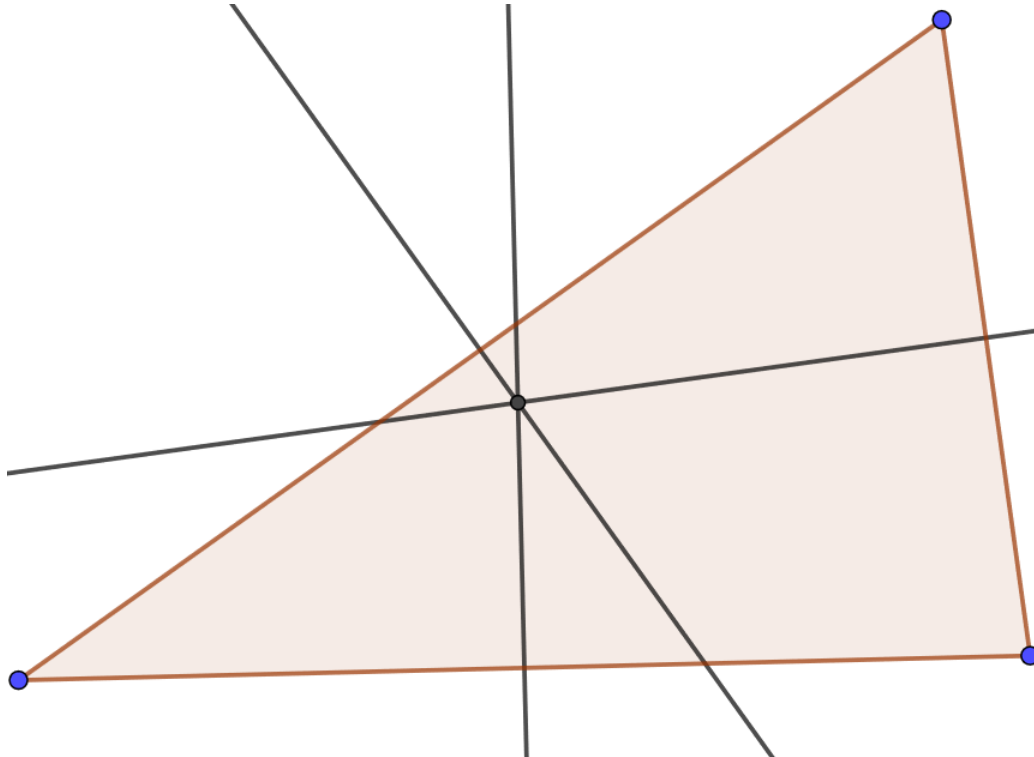


Figure 3.3.1: GeoGebra: Perpendicular Bisectors

Use the Geogebra example in 3.3.2 to experiment with the relationship of the three medians of a triangle. Move the vertices of the triangle around. What remains true about the medians?

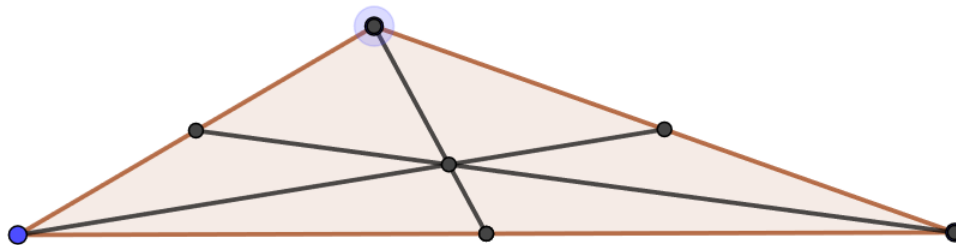


Figure 3.3.2: GeoGebra: Medians

Use the Geogebra example in 3.3.3 to experiment with the relationship of the three angle bisectors of a triangle. Move the vertices of the triangle around. What remains true about the angle bisectors?

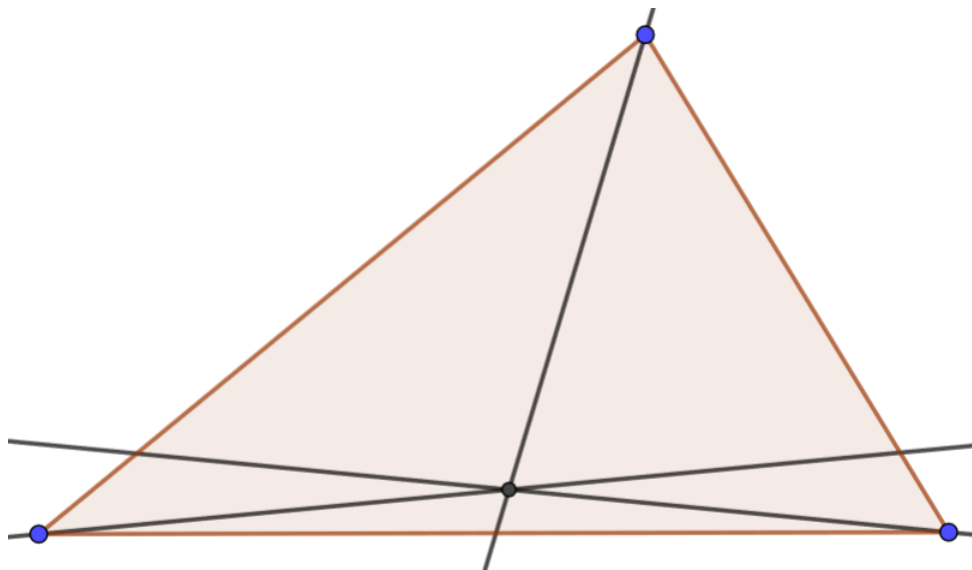


Figure 3.3.3: GeoGebra: Angle Bisectors

Use the Geogebra example in 3.3.4 to experiment with the relationship of the three altitudes of a triangle. Move the vertices of the triangle around. What remains true about the altitudes?

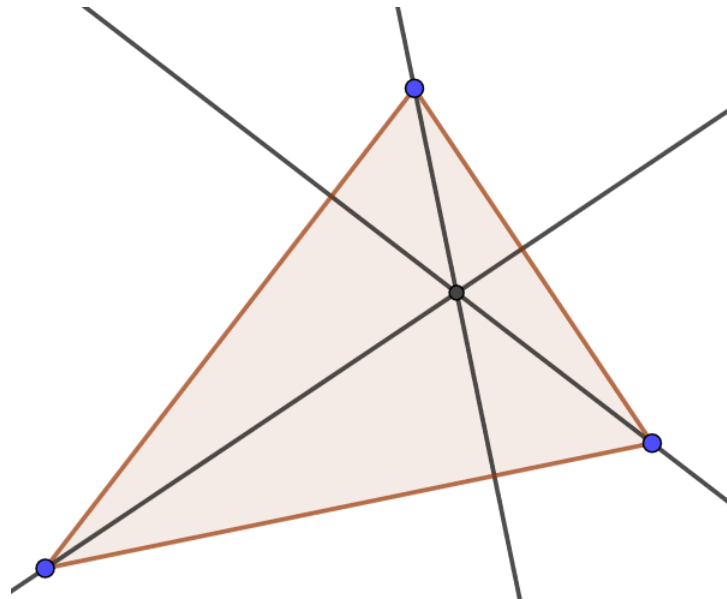


Figure 3.3.4: GeoGebra: Altitudes

Construct $\triangle ABC$. Construct $\triangle XYZ$ such that $X-B-Y$, $Y-C-Z$, $Z-A-X$ and $XY \parallel AC$, $YZ \parallel AB$, $ZX \parallel BC$. Construct the perpendicular bisectors of $\triangle XYZ$. What appears to be true of these with respect to $\triangle ABC$.

3.3.2 Prove

Lemma

Consider three points A, B, C with ℓ_1 and ℓ_2 the perpendicular bisectors of AB and BC respectively. Let $M_2 = \ell_2 \cap BC$. Show $\ell_1 \parallel \ell_2$ implies the existence of $D = \ell_2 \cap AB$ such that A, B and D are collinear and $\angle BDM_2$ is a right angle.

Theorem: Perpendicular bisectors

Prove the conjecture about the perpendicular bisectors.

Theorem: Circumcenter

Three points uniquely determine a circle.

Lemma

Two medians intersect at a point $\frac{2}{3}$ of the way down both medians.

Theorem: Medians

Prove the conjecture about the medians.

Theorem

A point is on the angle bisector of an angle if and only if it is equidistant from both sides of the angle.

Theorem: Angle bisectors

Prove the conjecture about the angle bisectors.

Theorem : Incenter

For each triangle there exists a circle inside and tangent to all three sides.

Theorem : Altitudes

Prove the conjecture about the altitudes.

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3.4: Constructions

For each construction figure out how to do it using the classic Greek tools: a straight edge and rusty compass (okay that isn't quite classic Greek). Note you can use the circle and line tools in Geogebra to perform these. Next prove that your construction works.

3.4.1 Discover and Prove Construction

1. Construct an equilateral triangle with side length matching a given segment.
 2. Given a line segment construct the perpendicular bisector of it.
 3. Construct a square with side length matching a given segment.
 4. Construct the midpoint of a line segment.
 5. Construct the bisector of given angle.
 6. Copy an angle.
 7. Construct a line parallel to a given line through a given point.
-

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CHAPTER OVERVIEW

4: Transformational Geometry

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4.1: Transformation

4.1.1 Planar Transformations

Definition: Transformation

A function is a **transformation** if and only if it is one-to-one and onto.

Definition: Planar Transformation

A transformation is a **planar** transformation if and only if it is from $\mathbb{R}^2$ to $\mathbb{R}^2$.

For this course all transformations will be transformations of the Euclidean plane.

Describe the effect of each of the following transformations by considering the effects on the region (area) with the following vertices. (0,0), (2,0), (3,5), (0,4). Hint: map the vertices, then map the lines connecting the vertices.

1. $T_1: (x, y) \rightarrow (y, x)$.
2. $T_2: (x, y) \rightarrow (x/2, y/2)$.
3. $T_3: (x, y) \rightarrow (1-x^3, 1-y^3)$
4. $T_4: (x, y) \rightarrow ((x^2+y^2)^{1/2}x, (x^2+y^2)^{1/2}y)$.

4.1.2 Isometry

Definition: Isometry

A transformation is an **isometry** if and only if $\|P - Q\| = \|T(P) - T(Q)\|$.

Determine which of the following transformations are isometries.

1. $T_1: (x, y) \rightarrow (2x, 2y)$
2. $T_1: (x, y) \rightarrow (-y, -x)$
3. $T_1: (x, y) \rightarrow \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Lemma

The composition of two isometries is an isometry.

Theorem: Isometries preserve colinearity

For any isometry T if A, B and C are colinear, then $T(A), T(B)$ and $T(C)$ are colinear.

Corollary: Isometries preserve betweenness

For any isometry T if $A-B-C$ then $T(A)-T(B)-T(C)$.

Theorem: Isometries preserve triangles

For any isometry T and any three points $\triangle ABC \cong \triangle T(A)T(B)T(C)$.

Theorem: Isometries preserve angles

For any isometry T $m\angle ABC = m\angle T(A)T(B)T(C)$ and any three points.

Theorem: Isometries preserve parallelism

For any isometry if $T \ell_1 \parallel \ell_2$ and only if $T(\ell_1) \parallel T(\ell_2)$.

Theorem: Isometries preserve circles

For any isometry T circles are mapped to congruent circles.

4.1.3 Dilations

Definition: Dilation

A transformation is a **dilation** if and only if it can be defined by a point Z and a ratio k such that $T(P)=Q$ where $Z-P-Q$ and $\|ZQ\|/\|ZP\|=k$.

Definition: Similarity

A transformation is a **similarity** if and only if it can be expressed as a composition of an isometry and a dilation.

Lemma: Similarity scales segments uniformly

For any similarity T , $\|T(A)T(B)\|/\|AB\|=k$.

Theorem: Similarity preserves colinearity

For any similarity T if A , B , and C are colinear, then $T(A)$, $T(B)$, and $T(C)$ are colinear.

Corollary: Similarity preserves betweenness

For any similarity T if then $A-B-C$ then $T(A)-T(B)-T(C)$.

Theorem: Similarity preserves triangle similarity

For any similarity T and any three points $\triangle ABC \sim \triangle T(A)T(B)T(C)$.

Theorem: Similarity preserves angles

T For any similarity and any three points $\angle mABC \sim \angle mT(A)T(B)T(C)$.

Theorem: Similarity preserves parallelism

For any similarity T $\ell_1 \parallel \ell_2$ if and only if $T(\ell_1) \parallel T(\ell_2)$.

Theorem

For any similarity T circles are mapped to circles.

Lemma: Similarities are closed

The composition of two similarities is a similarity.

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4.2: Analytic Transformational Geometry

The goal is to develop matrix formulas for arbitrary isometries using the basic isometry formulas given below as building blocks.

Definition: Homogeneous coordinates

A point with normal coordinates (x, y) in homogeneous coordinates is written $(x, y, 1)$.

Table 4.2.1: Linear Transformations for Isometries

Translate	$T(x, y, 1) = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	
Reflect over the y-axis	$M_y(x, y, 1) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	(4.2.1)
Reflect over the x-axis	$M_x(x, y, 1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	(4.2.2)
Rotate counterclockwise about the origin	$R_\phi(x, y, 1) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	(4.2.3)

Goal: develop a rotation about a point $[x_0, y_0]^T$ using the following steps.

1. Find a transformation that moves $[x_0, y_0]^T$ to the origin.
2. Find a transformation that moves $[x_0, y_0]^T$ to the origin then rotates by ϕ .
3. Find a transformation that moves $[x_0, y_0]^T$ to the origin, rotates by ϕ , then returns the origin to $[x_0, y_0]^T$.
4. State, using matrix notation, a transformation that rotates the plane about a point $[x_0, y_0]^T$ by ϕ .

Goal: develop a reflection about a vertical line given by $x=a$ using the following steps.

1. Find a transformation that move the line $x=a$ to the y-axis.
2. Find a transformation that move the line $x=a$ to the y-axis, then reflects the plane over the y-axis.
3. Find a transformation that move the line $x=a$ to the y-axis, reflects the plane over the y-axis, then returns the y-axis to the line $x=a$.
4. State, using matrix notation, a transformation that reflects about an arbitrary vertical line $x=a$.

Goal: develop a reflection about a horizontal line given by $y=b$ using the following steps.

1. Find a transformation that move the line $y=b$ to the -axis.

2. Find a transformation that move the line $y=b$ to the x -axis, then reflects the plane over the x -axis.
3. Find a transformation that move the line $y=b$ to the x -axis, reflects the plane over the x -axis, then returns the x -axis to the line $y=b$.
4. State, using matrix notation, a transformation that reflects about an arbitrary horizontal line $y=b$.

Develop a reflection about an arbitrary (non-vertical) line.

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4.3: Algebra of Transformations

4.3.1 Explore

Definition: Translation

An transformation T is a translation if and only if there exists a non-zero constant vector $\mathbf{v}$ such that $T(P) - P = \mathbf{v}$ for all points P .

Definition: Rotation

An transformation T is a rotation if and only if there exists a fixed point C and constant angle α such that $m\angle PCT(P) = \alpha$ and $\|CP\| = \|CT(P)\|$ for all points P .

Definition: Reflection

An transformation T is a reflection if and only if there exists a fixed line ℓ such that the line perpendicular to ℓ through P contains $T(P)$ and the distances from P and $T(P)$ to the ℓ are equal.

Theorem

Translations, rotations, and reflections are isometries.

Draw an arbitrary triangle $\triangle ABC$. Draw the result $\triangle A'B'C'$ of some translation. Draw the result $\triangle A''B''C''$ of some translation applied to $\triangle A'B'C'$. Determine which type of isometry would transform $\triangle ABC$ to $\triangle A''B''C''$.

Complete the following table of composition of isometries.

	Translate	Reflect	Rotate
Translate			
Reflect			
Rotate			

4.3.2 Prove

How many isometry types are there?

How many isometry types are needed to generate all isometry types?

How many isometries are needed to generate all isometries?

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4.4: Symmetries

4.4.1 Explore Symmetries

Definition: Symmetry

A set of points A has symmetry of type T for some transformation T if and only if $T(A) = A$.

Confirm $T(x, y) = (-y, x)$ is a symmetry of the set $\{(1, 1), (-1, 1), (-1, -1), (1, -1)\}$.

Demonstrate $T(x, y) = (-y, x)$ is not a symmetry of the set $\{(0, 0), (1, 0), (1, 1), (0, 1)\}$.

What points do you have to add to $\{(1, 1), (-1, 1), (-1, -1), (1, -1), (1, 0)\}$ so $T(x, y) = (-y, x)$ is a symmetry of this set?

List all symmetries of a square by labeling the vertices and giving the type and parameters for the transformations.

List all symmetries of a regular n -sided polygon (n -gon) by labeling the vertices and giving the type and parameters for the transformations.

For one of the regular n -gons check the following.

1. What is the composition of two rotational symmetries?
2. What is the composition of two reflection symmetries?
3. What is the smallest number of symmetries you can use to generate all the symmetries?

Draw some regular n -gon. Color in the n -gon so that the colored figure maintains the rotational symmetries, but not the reflectional symmetries.

Draw some regular n -gon. Color in the n -gon so that the colored figure maintains the reflectional symmetries, but not the rotational symmetries.

Draw a figure that has translational symmetry.

Draw a figure that has translational symmetry and exactly one reflectional symmetry.

Draw a figure that has translational symmetry and rotational symmetry.

Draw a figure that has dilational symmetry.

4.4.2 Explore Tessellations

Definition: Tessellation

A covering of the plane is a **tessellation** if and only if it consists of a single shape infinitely reproduced using a finite set of transformations.

Definition: Tiling

A covering of the plane is a **tiling** if and only if it consists of a finite set of shapes infinitely reproduced using a finite set of transformations.

Analyze the tessellation as follows.

1. Identify the generating shape.
2. Identify the smallest set of transformations that can generate the tessellation.
3. List all symmetries of the tessellation.
4. Identify the smallest set of symmetries of the tessellation that can generate all the symmetries of the tessellation.

Definition: Conway Label

The following labeling of tessellations derives from the book *The Symmetries of Things* by John Conway, Heidi Burgiel, and Chaim Goodman-Strauss. Follow these steps in order to identify and label the type of symmetry group of a tessellation. The resulting notation is called the *signature*

1. Identify all lines of reflection.
2. If two or more lines of reflection intersect in a point, write $*n_1n_2\ldots$ where n_1, n_2 are the number of lines intersecting at each unique point of intersection.
3. If any line of reflection does not intersect other lines of reflection, just write one $*$ for each of these.
4. Identify any rotations that are not the composition of reflections already listed.
5. Write $n_1n_2\ldots$ in front of any $*$ for each rotation where n_1, n_2 are the order of the rotations.
6. Identify any glide reflections that are not the composition of reflections or rotations already listed.
7. Write $\times$ at the end of the signature for each of these glide reflections.
8. Identify any translations that are not the composition of other symmetries already listed.
9. Write $\circ$ at the front of the signature for each pair of these translations.

See the example signatures in Figures 4.4.1 to 4.4.4.

Find the signatures of two tessellations from the class archive at [here](#). You may not choose two with the same signature.

Find the signature of the tessellation in Figure 4.4.1.

Begin the tessellation project.

4.4.3 Tessellation Images

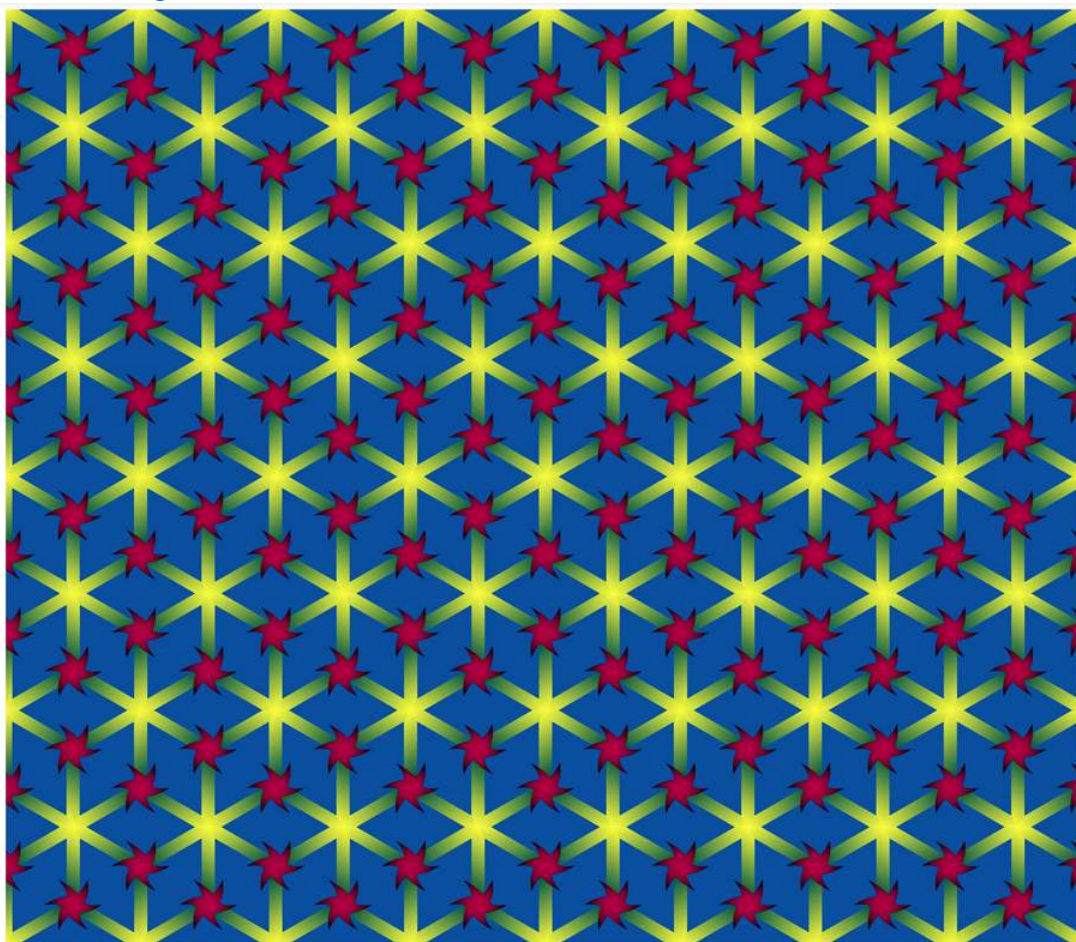


Figure 4.4.1: Tessellation

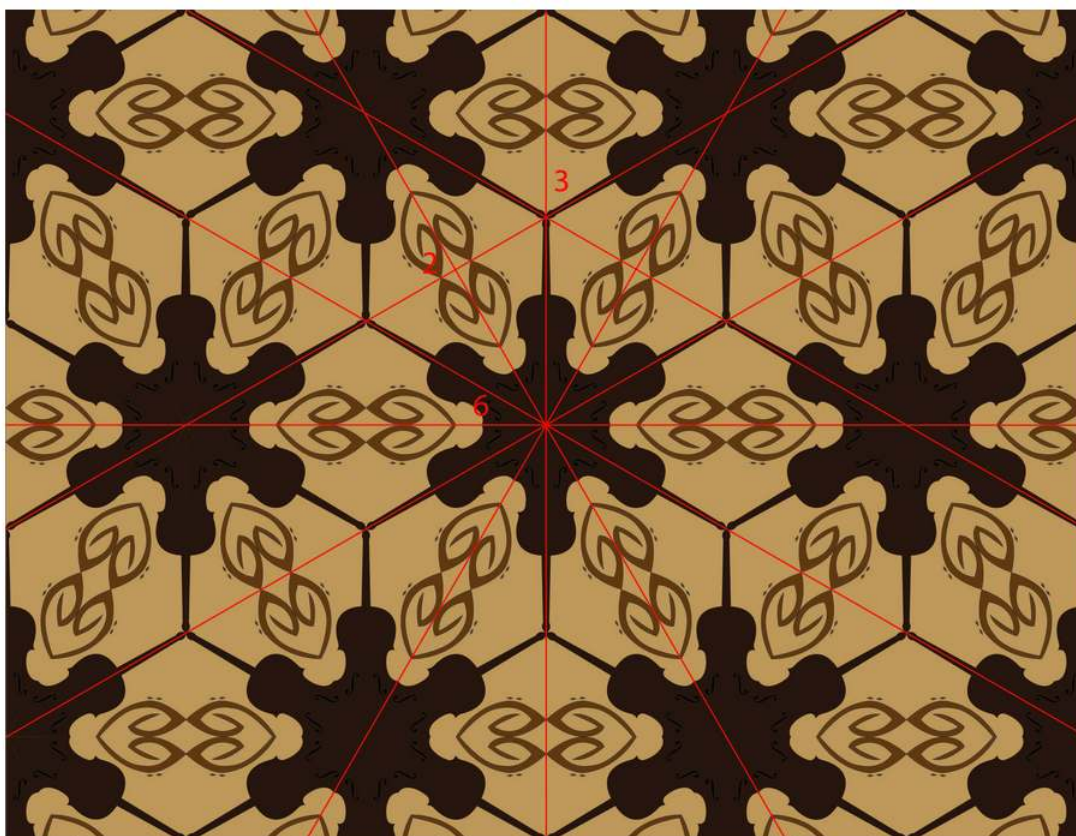


Figure 4.4.2: Signature *236 (Variation on a Theme by Scott Roseman)

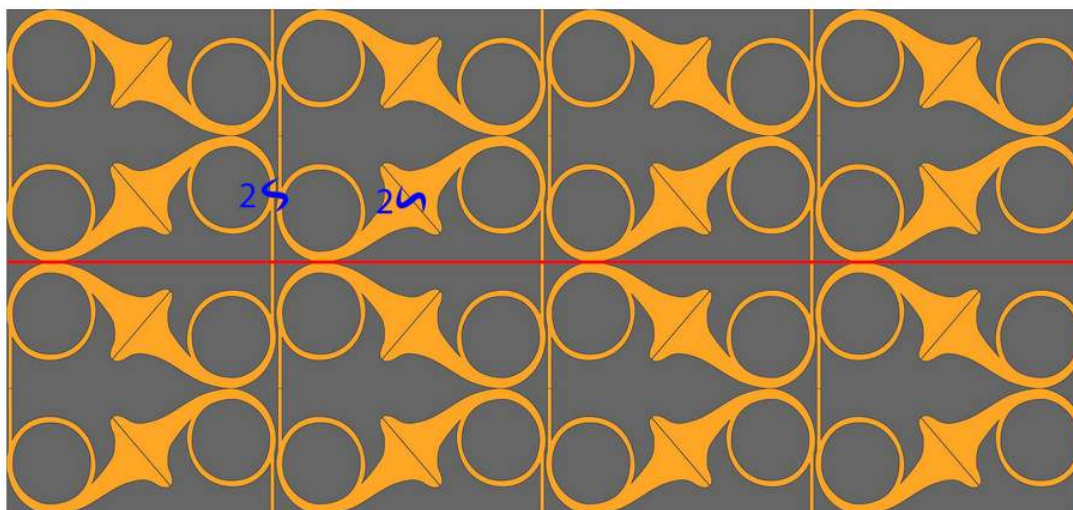


Figure 4.4.3: Signature 22*

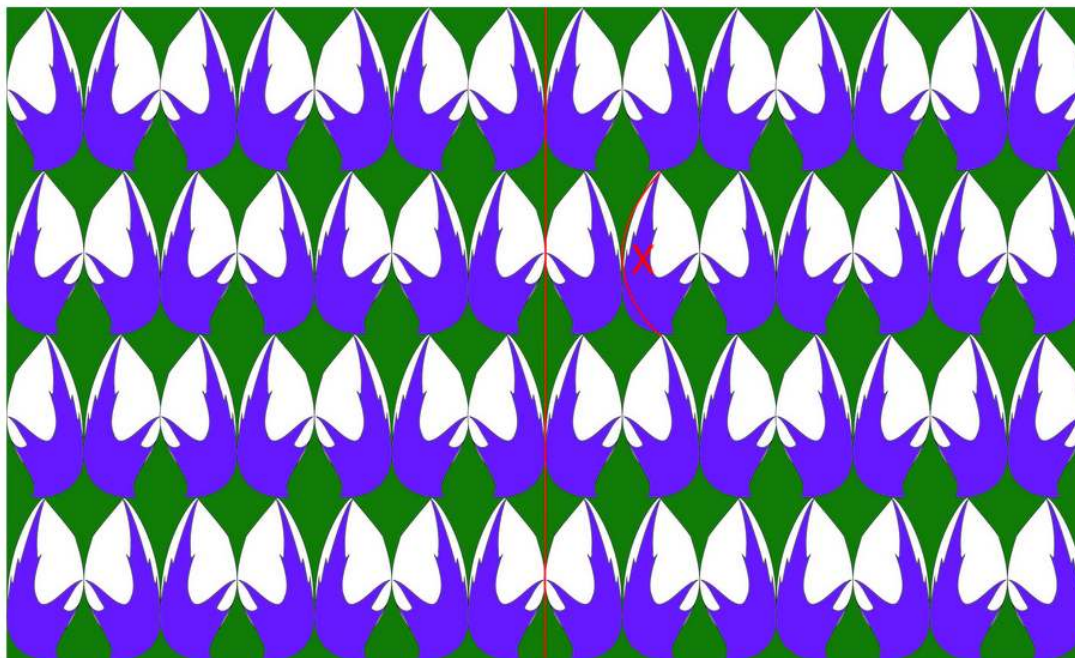


Figure 4.4.4: Signature *X

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CHAPTER OVERVIEW

5: Hyperbolic Geometry

[5.1: Hyperbolic Geometry](#)

[5.2: A Model for Hyperbolic Geometry](#)

[5.3: Theorems of Hyperbolic Geometry](#)

[5.4: Parallels in Hyperbolic](#)

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5.1: Hyperbolic Geometry

Hyperbolic geometry results by replacing the Euclidean parallel postulate with the following.

Axiom

Given a line and a point not on that line there exists at least two lines through the point and parallel to the lines.

There were three major variants (wordings) of the Euclidean parallel postulate. Conjecture what these look like in hyperbolic geometry.

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5.2: A Model for Hyperbolic Geometry

Hyperbolic geometry can be drawn with the aid of the Poincaré disc model. The hyperbolic plane is represented by a disc with the border not included ("open disc" in analysis terms). Lines are either diameters or circular arc that are orthogonal to the disc. The origin is the center of the disc. You will use the provided Geogebra file or one you search for online in Geogebra with special tools to explore hyperbolic geometry.

Construct each of the following hyperbolic figures.

1. Triangle using at least two lines that are diameters.
2. Triangle using exactly one line that is a diameter.
3. Triangle using no lines that are diameters.
4. Quadrilateral (Can you make it a square?)

Explore parallelism in hyperbolic geometry.

1. Construct a line and select a point not on that line. Construct two lines through that point parallel to the given line.
2. How many lines through that point parallel to the given line could be constructed?
3. Do any of these parallel lines have special properties? Properties might be easier to describe in terms of the model.
4. Construct two parallel lines. For ease make them large and close in the model. What seems to be true about the distance between the parallel lines?

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5.3: Theorems of Hyperbolic Geometry

Note when working on these problems the model may be useful in figuring out why something works. However the proofs should be directly based on the axioms. It is legitimate to prove a statement using the model description if the model has been proven to be equivalent to the axioms. For this course proofs using the model will be worth fewer points than proofs directly from the axioms.

Lemma: Hyperbolic Parallel Corollary

Given a line and a point not on that line there exists infinitely many lines through that point parallel to the given line.

Lemma: Sensed Parallel

Given a line and a point not on that line there exists a first line through that point parallel to the given line.

Definition: Sensed Parallel

A line is a *sensed parallel* if and only if it is the first line parallel to a given line on that side through a given point.

Definition: Angle of Parallelism

The smaller angle formed by a sensed parallel and a transversal through the given point is the *angle of parallelism* if and only if the transversal is perpendicular to the given line.

Corollary: Term

The angle of parallelism is the same on the left and right.

Theorem

The angle of parallelism is less than $\pi/2$.

Lemma

Consider the following illustrated in 5.3.1. ℓ is the right sensed parallel to n at P . Let S be on ℓ to the left of P . Suppose line m through S is the sensed parallel to n at S . Show that if T is on m to the left of S then must be below TP to the right of T . Further if U on ℓ such that $U-S-P$ and $A=TP \cap n$, then sensed parallel must be above SA to the right of A and above UA .

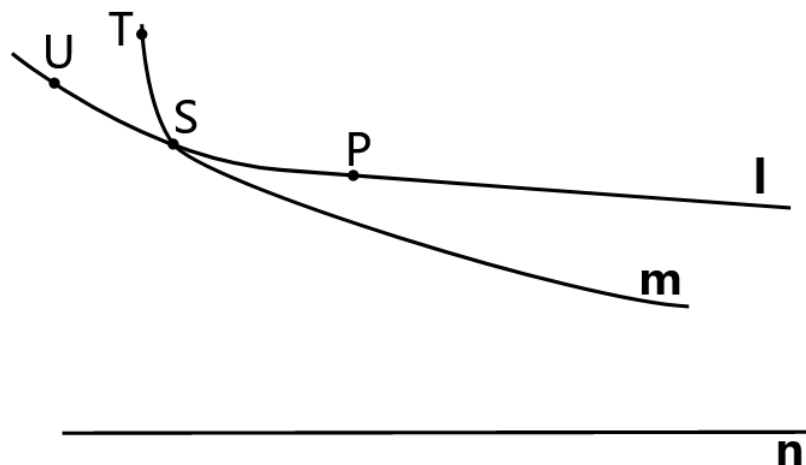


Figure 5.3.1: Sensed Parallels Left

Lemma: Term

Consider the following illustrated in 5.3.2. ℓ is the right sensed parallel to n at P . Let S be on ℓ to the right of P . Suppose line m through S is the sensed parallel to n at S . Show that if U and T are on such that $U-S-T$ and $A=TP \cap n$ then m is below UA to the right of U and above PA .

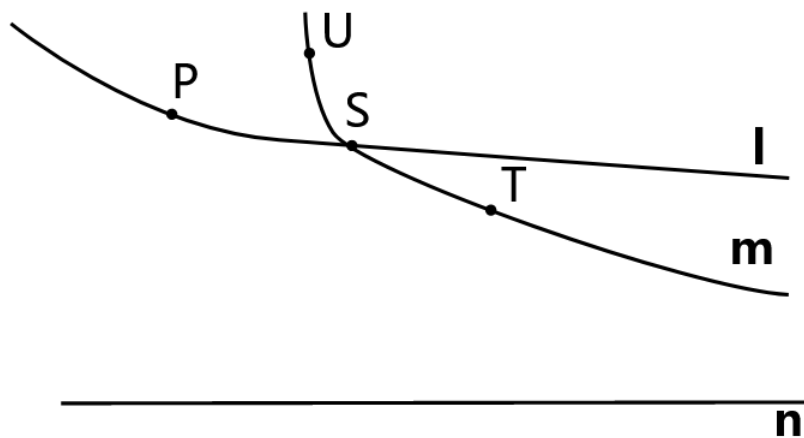


Figure 5.3.2: Sensed Parallels Right

Lemma

If ℓ is the right sensed parallel to m at P , then ℓ is the right sensed parallel to m at any point to the left of P on ℓ .

Theorem: Sensed Everywhere

If ℓ is the sensed parallel to m at a point P , then ℓ is the sensed parallel to m at any point Q also on ℓ .

Definition: Omega Triangle

A pair of parallel lines and a transversal is a **omega triangle** if and only if the parallels are sensed parallels.

Draw an omega triangle using the Poincaré disc model.

Lemma: Crossbar for Omega Triangles

If a line k contains a vertex and an interior point of an omega triangle, it intersects the side opposite the vertex.

Does a version of Pasch's Axiom work for omega triangles? Let ℓ and m be sensed parallels. Let AB be a transversal with $A \in \ell$ and $B \in m$. Let M be the midpoint of AB and D be the foot of the perpendicular from M to ℓ . Also choose $F \in m$ on the opposite side of AB from D such that $BF \cong AD$. Let $C \in \ell$ be such that $C-A-D$. Prove that $\angle CAB \neq \angle AB\omega$.

Lemma: Omega Triangle Exterior Angle

An exterior angle of an omega triangle is greater than the opposite interior angle.

Lemma: Angle-Side Congruency

Two omega triangles are congruent if the length of the finite sides and the measure of one pair of corresponding angles are congruent.

 Lemma: Angle-Angle Congruency

Two omega triangles are congruent if corresponding pairs of angles are congruent.

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5.4: Parallels in Hyperbolic

Theorem

If two lines are sensed parallels to a third line, they are also sensed parallels to each other.

Theorem

If two lines are sensed parallels to a third line, the line farthest away has the smallest angle of parallelism.

Definition: Saccheri Quadrilateral

A quadrilateral is a **Saccheri quadrilateral** if and only if it has two consecutive right angles adjacent to two congruent sides. The side orthogonal to two sides is the base. The opposite side is the summit.

Theorem

The non-right angles in a Saccheri quadrilateral are congruent.

Theorem

The line segment joining the midpoint of the base to the midpoint of the summit is orthogonal to both.

Lemma

Let ABCD be a Saccheri quadrilateral with right angles at A and B. Prove that $\angle AD\Omega \cong \angle BC\Omega$.

Theorem

The non-right angles in a Saccheri quadrilateral are acute.

Theorem

Parallel lines are not everywhere equidistant.

Theorem

A transversal perpendicular to two parallel lines is unique.

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CHAPTER OVERVIEW

6: Projective Geometry

[6.1: Axioms for Projective Geometry](#)

[6.2: Perspectivities](#)

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6.1: Axioms for Projective Geometry

6.1.1 Motivating Illustration

Consider the following illustration as a motivation of this geometry. Consider standing in the middle of Kansas looking down a perfectly straight road that extends all the way to the horizon.

1. Presuming perfect construction, the two sides of the road are lines with what geometric property?
2. As you look toward the horizon, what do the sides of the road appear to do?
3. Two lines always intersect in one what?
4. Consider all the lane line markings (there are more than two). All of these lines are what compared to each other and appear to do what?
5. If you are in an intersection of two roads (not in the same direction), will the lane markings all converge together?
6. How many different convergent locations are there?

Definition: Ideal Point

A point is an **ideal point** if and only if it is the intersection of parallel lines. These are sometimes called "points at infinity."

Definition: Ideal Line

A line is the **ideal line** if and only if it consists of solely ideal points.

6.1.2 Axioms for Projective Geometry

Axiom: Projective Geometry

1. A line lies on at least two points.
2. Any two distinct points have exactly one line in common.
3. Any two distinct lines have at least one point in common.
4. There is a set of four distinct points no three of which are colinear.
5. All but one point of every line can be put in one-to-one correspondence with the real numbers.

The first four axioms above are the definition of a finite projective geometry. The fifth axiom is added for infinite projective geometries and may not be used for proofs of finite projective geometries.

Theorem

A line lies on at least three points.

Theorem

Any two, distinct lines have exactly one point in common.

Lemma

For any two distinct lines there exists a point not on either line.

Theorem

There exists a one-to-one correspondence between the points of any two lines.

 Theorem

Every point lies on the same number of lines.

 Corollary

A projective plane in which each line lies on exactly $k+1$ points has a total of k^2+k+1 points and k^2+k+1 lines.

6.1.3 Duality

 Definition: Projective Duality

A statement is the **projective dual** of another statement if and only if one statement is obtained from the other by switching the roles of "point" and "line."

 Theorem

Each point is incident with at least three lines.

 Theorem

There exist four lines no three of which are coincident in a point.

 Theorem

There is a one-to-one correspondence between the real numbers and all but one of the lines incident with a point.

 Theorem: Projective Duality

The projective dual of every projective theorem is also true.

 Theorem

Every line consists of the same number of points.

 Theorem

There exists a one-to-one correspondence between the lines thru any two points.

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6.2: Perspectivities

Review the motivation for projective geometry and answer the following.

1. If you view a tree from 10 meters away and 100 meters away (viewing from the same side), what is the difference in the objects?
2. As you continue to move away from the tree, toward what does the mapping of the tree approach?
3. Look up the term 'vanishing point' in an art book. Explain its relationship to this.

Definition: Perspectivity

A transformation is a **perspectivity** if and only if it maps the points of a line to the points of another line such that all lines from points to their images are incident in a single point.

Draw two distinct lines. Choose a point not on either line. Choose three points on one of the lines and find the points on the second line to which they are mapped by the perspectivity defined by your chosen point.

Investigate the composition of perspectivities as follows.

1. Draw three distinct lines ℓ_1, ℓ_2, ℓ_3 .
2. Choose a point A not on any of the lines.
3. Select three points P_1, Q_1, R_1 on ℓ_1 and map them to P_2, Q_2, R_2 ON ℓ_2 by the perspectivity defined by A .
4. Choose a point distinct from all other points.
5. Map P_2, Q_2, R_2 to P_3, Q_3, R_3 on ℓ_3 by the perspectivity defined by B .
6. Determine if P_1, Q_1, R_1 are mapped to P_3, Q_3, R_3 by any perspectivity.

Definition: Projectivity

A transformation is a **Projectivity** if and only if it can be written as a composition of perspectivities.

Definition: Triangle

A set of points is a **triangle** if and only if it is size three.

Definition: Trilateral

A set of lines is a **trilateral** if and only if it is size three.

Definition: Perspective from a point

Two triangles are **perspective with respect to a point** if and only if the lines connecting corresponding pairs of vertices are incident in a point.

Definition: Perspective from a line

Two trilaterals are **perspective with respect to a line** if and only if the corresponding sides are incident on a line.

Theorem: Desargues Theorem

Triangles perspective from a point are perspective from a line.

Theorem: Fundamental Theorem of Projective Geometry

A projectivity is uniquely defined by three points and their images.

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